NIDHIFLOW | ■■■■ ■■■■

Voice-confirmed, tamper-evident community passbook for Kerala Self-Help Groups

THE PROBLEM

Many community savings groups still depend on paper passbooks or a single person to record weekly savings, loans and repayments. These workflows can be difficult to audit, create barriers for members who are less comfortable with text-heavy apps, and make incorrect or changed entries hard to notice. Voice recognition can improve accessibility, but an unverified transcription should never be trusted to record a financial transaction on its own.

OUR SOLUTION

NidhiFlow is a Malayalam-first, voice-assisted digital passbook concept for Kudumbashree and other grassroots Self-Help Groups. It uses Gnani.ai speech-to-text as an input aid, then places human confirmation and group-governance checks between speech recognition and ledger entry. The application is designed to make community records easier to create, review and audit without allowing AI to independently approve financial activity.

HOW IT WORKS

1. Capture A member's spoken Malayalam or Manglish entry is recorded and its audio is retained as evidence.	**2. Cross-check** Two transcription formats are compared; mismatches and uncertain details are flagged for review.
3. Verify The app reads the proposed entry back in Malayalam for people to confirm, with officer co-signing and accounting rules where required.	**4. Preserve** Confirmed entries are linked using SHA-256 hashes, allowing later checks to reveal changes to ledger records or stored audio.

WHAT MAKES IT DIFFERENT

- **Malayalam voice-first interaction:** supports spoken Malayalam and code-mixed input.
- **Human-in-the-loop:** audible read-back and confirmation keep people responsible for entries.
- **Governance guardrails:** checks include two-person loan confirmation, conflict-of-interest prevention and repayment limits.
- **Evidence-linked integrity:** audio hashes and a SHA-256 hash-chained ledger support tamper detection and audit review.

TECHNOLOGY

Python · Streamlit · Gnani.ai speech-to-text · SQLite · SHA-256

PURPOSE & CURRENT SCOPE

NidhiFlow aims to help SHG members and office-bearers maintain clearer, more accessible and more accountable community records. It is a prototype: speech recognition can make mistakes, and hash chaining is tamper-evident rather than a guarantee that a locally controlled database cannot be replaced. Human review remains essential.

Making community savings easier to record, understand and verify.